ONKAR PURJALKAR

Machine Learning Engineer & Software Developer

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PROFESSIONAL SUMMARY

Passionate B.Tech student and aspiring Machine Learning Engineer & Software Developer with hands-on experience building intelligent applications, predictive systems, and conversational agents. Proficient in Python, Machine Learning, data science and full-stack web development. Proven capability in developing end-to-end applications from model engineering to interactive UI dashboards, with practical internship experience in software quality assurance.

EDUCATION

Bachelor of Technology (B.Tech)-MGM's college of Engineering , Nanded **Expected Graduation: 2028**
Major in Information Technology & Minor in Artificial Intelligence & Machine learning

Diploma in Information Technology **Completed: 2025**
Government Polytechnic, Nanded

WORK EXPERIENCE

JOBITO — Software Quality Assurance (SQA) Intern **2 Months**

- Executed manual testing and developed test plans to ensure functionality, reliability, and security of web applications.
- Collaborated with core developer teams to document, track, and verify resolution of bugs in a database-driven system.
- Gained hands-on insights into software development life cycles (SDLC) and QA best practices.

PROJECTS

AIVICS — AI-Enhanced Virtual Interaction & Control System **Python, OpenCV, Streamlit, APIs**

- Engineered an AI-powered desktop virtual assistant with voice activation, computer vision gesture control, and automation features.
- Implemented **Hotword Detection** for hands-free wake commands and **Air Canvas** using hand-tracking and OpenCV for in-air writing.
- Integrated **Eye Mouse Control** via face landmarks, voice search activation, automatic system volume controls, and network speed testing.
- Designed automated messaging systems triggering WhatsApp, Email, SMS via python-based libraries and APIs.

Air Quality Monitoring & Predictor **Random Forest, Streamlit, Pandas**

- Built an end-to-end machine learning system using **Random Forest Regression** to forecast historical and future Air Quality Index (AQI) values.
- Developed an interactive web interface using **Streamlit** to plot real-time pollution metrics, trends, and generate automated hazard warnings.

- Preprocessed complex environmental datasets using NumPy and Pandas to optimize model accuracy and training efficiency.

Multiagent Chatbot System **Python, LLMs, Streamlit, NLP**

- Developed a conversational AI architecture where multiple specialized LLM agents collaborate to resolve user queries.
- Implemented task-routing algorithms separating research assistance, general chat agents.

TECHNICAL SKILLS

Programming & Web:	Python, PHP, JavaScript, SQL, HTML5, CSS3
Machine Learning & Data:	Random Forest, Scikit-Learn, NumPy, Pandas, OpenCV, NLP
Frameworks & Databases:	Streamlit, Flask, MySQL, SQLite
Tools & Concepts:	Git & GitHub, API Integrations, Software Testing (QA), System Automation

CERTIFICATIONS

- Machine Learning with Python — **IBM**
- Python, NumPy and Pandas — **Infosys Springboard**
- The Complete Full Stack Web Development Bootcamp — **Udemy**
- PHP and MySQL — **Spoken Tutorial**